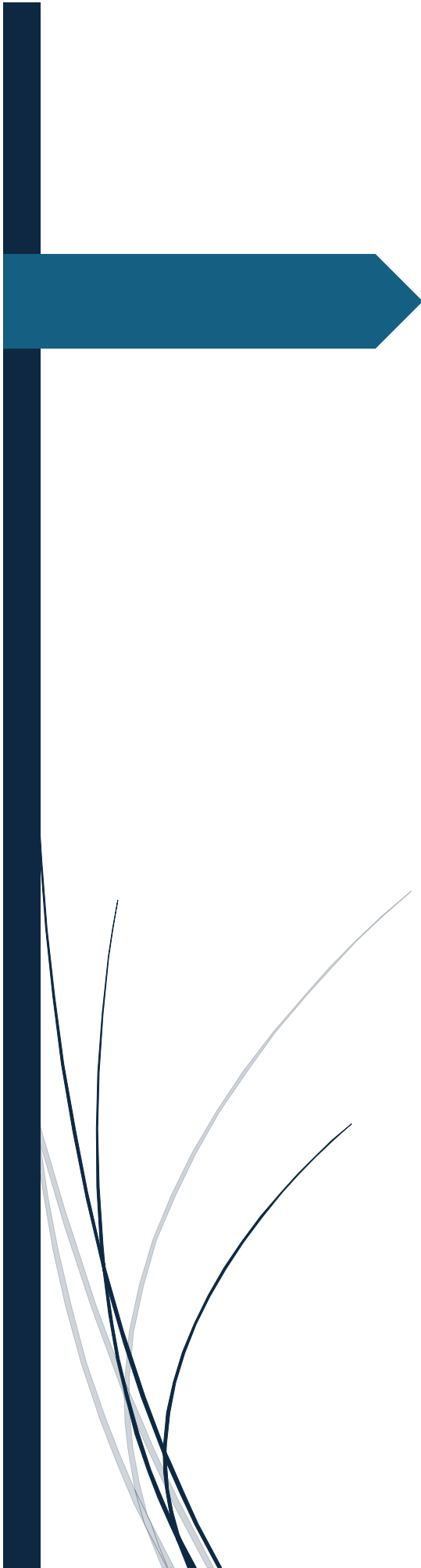




Assignment 4

Component Specific
Troubleshooting

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

➤ **RAM Reseating Steps:**

1. Shut down the laptop and unplug the charger.
2. Open the back cover.
3. Locate the RAM slot.
4. Remove the RAM by releasing the side clips.
5. Check for dust or damage.
6. Re-insert the RAM firmly into the slot.
7. Close the cover and turn on the laptop.
8. Check if the laptop boots normally.

Issue observed: No damage found. RAM was reseated successfully and the laptop booted normally.

2. **Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Hint: Think about display issues, driver checks, and physical connections.**

➤ **GPU Troubleshooting Checklist:**

1. **Identify symptoms:** No display, screen flickering, crashes, or low FPS.
2. **Check connections:** Make sure the GPU is properly seated and power cables are connected.
3. **Check monitor:** Test the monitor and HDMI/DisplayPort cable.
4. **Check drivers:** Update or reinstall the GPU driver.
5. **Check temperature:** Make sure the GPU is not overheating.
6. **Test the fix:** Restart the PC and run a game/test to confirm the GPU works normally.

3. Simulate a storage issue by unplugging and re-plugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

➤ **Storage Issue Simulation:**

1. **Before unplugging:** BIOS detects the SSD/HDD normally.
2. **Unplug the drive:** BIOS shows “**No Storage Device Detected.**”
3. **Re-plug the drive:** Connect the SATA/NVMe drive properly.
4. **After reconnecting:** BIOS detects the drive again.
5. **Result:** The system boots normally and the drive is accessible.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

- **Check RAM first** because faulty or poorly connected RAM can cause random restarts.

Diagnostic steps:

1. Shut down the PC.
2. Remove and reseal the RAM.
3. Check for dust or damage.
4. Test one RAM module at a time.
5. Run **Windows Memory Diagnostic**.
6. Restart the PC and check if the problem is fixed.